Dohee Kim, Ph. D.

Assistant Professor
Department of Artificial Intelligence Engineering
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Research Interest

Time-series Analysis, Deep Learning, Feature Engineering, Multi-modal AI, Large Language Models

Education

Mar. 2019 ~ Feb. 2024 — **Pusan National University** (GPA 4.2 / 4.5) — Busan, Korea
Ph.D. in Industrial Engineering
Thesis: Deep Learning Models for Short-term and Long-term Time Series Forecasting (Advisor: Hyerim Bae)

Mar. 2012 ~ Feb. 2019 — **Pusan National University** — Busan, Korea
B.S. in Industrial Engineering

Work Experience

Sep. 2026~ Present — **Design thinking SW Center, SW-Centered University Project Group, Changwon National University**
Director

Aug. 2026~ Feb. 2030 — **M.AX Center for Regional Strategic Industries, RISE Project Group Changwon National University**
Director

Jul. 2026~ Present — **AX Center, Department of Research Innovation, SMR^2 Platform National Research Laboratory, Changwon National University**
Head

Mar. 2026 ~ Feb. 2031 — **DRIVE Bootcamp Project Group, Changwon National University**
Deputy Director

Dec. 2025 ~ Present — **Department of Artificial Intelligence Engineering, Changwon National University**
Assistant Professor

Oct. 2024 ~ Nov. 2025 — **Safe & Clean Supply Chain Research Center, Pusan National University**
Research Professor

Sep. 2024 ~ Dec. 2024 — **College of Computing, Georgia Institute of Technology**
Visiting Scholar

Mar. 2024 ~ Sep. 2024 **Safe & Clean Supply Chain Research Center, Pusan National University**
Postdoctoral Researcher

Mar. 2024 ~ Sep. 2024 **Department of Industrial Engineering, Pusan National University**
Instructor

Teaching Experience

Fall, 2026 **Changwon National University**
Instructor, Graduate class – Database and Big Data Systems (AIB2005)
- All tasks such as Lecturing, Grading, Practical class, and Proctoring.

Fall, 2026 **Changwon National University**
Instructor, Undergraduate class – Python Programming in the Age of AI (GEA8810)
- All tasks such as Lecturing, Grading, Practical class, and Proctoring.

Spring, 2026 **Changwon National University**
Instructor, Graduate class – Multimodal AI (AIB3020)
- All tasks such as Lecturing, Grading, Practical class, and Proctoring.

Spring, 2026 **Changwon National University**
Instructor, Undergraduate class – Data Structures and Algorithm (AIA1007)
- All tasks such as Lecturing, Grading, Practical class, and Proctoring.

Fall, 2025 **Pusan National University**
Instructor, Graduate class – Shipping & Port Application (DS9700019)
- All tasks such as Lecturing, Grading, Practical class, and Proctoring.

Spring, 2024 **Pusan National University**
Instructor, Undergraduate class - Database (IE3400078-066)
- All tasks such as Lecturing, Grading, Practical class, and Proctoring.

Spring, 2024 **Pusan National University**
Instructor (Team teaching with Prof. Hyerim Bae),
Undergraduate class - Data Structure and Algorithm (IE1600795-066)
- Practical class, Grading, Proctoring.

Fall, 2022 **Pusan National University**
Teaching Assistant, Undergraduate class – Industrial Data Science (IE3500627-066)
- Practical class, Grading, Proctoring.
- Instructor: Hyerim Bae

Awards and Honors

2025 **Top viewed article WILEY 2025,** Journal of Forecasting

Jul. 2025 **Certificate of Commendation from the Mayor of Busan**, Busan Metropolitan City

Jan. 2024 **BK21's Best Paper Award**, 2023 BK21's Graduate Student Outstanding Papers Award, Pusan National University

Feb. 2019 **3rd Prize**, Undergraduate Capstone Design Term Project in Industrial Engineering, Pusan National University

Academic and Professional Services

Organizing Committee (Registration Chair), KOREA DATABASE CONFERENCE 2026(November 13-14)

Special Session Organizer, 'AI for Safety, Sustainability, and Automation in Smart Maritime Transport and Logistics Systems', IEEE CASE 2026

Session Chair, Korea Computer Congress 2026 (KCC 2026) (June 25)

Session Chair, 2026 KIIE Spring Conference (June 4-5)

Organizing Committee (Web Chair), Data Intelligence Workshop 2026(April 26-27)

Board Member, KIISE Data Society of Korea (KDS)

Program Committee Members and Reviewers, The 12th International Conference on Time Series and Forecasting (ITISE 2026)

Session Chair, 'Sustainable Maritime Logistics: Innovations for Safety, Efficiency, and Clean Port', ICICIC2025

Session Chair, 'Sustainable Maritime Logistics: Innovations for Safety, Efficiency, and Clean Port', ICICIC2024

Committee (Main Assistant), The 11th International Conference on Logistics and Maritime Systems (LOGMS 2023)

Editorial Board Member, JUTI: Jurnal Ilmiah Teknologi Informasi (Scientific Journal of Information technology), ITS, Indonesia

Peer Reviewer, IEEE Transactions on Big Data, Pattern Recognition, Scientific Reports, etc.

PUBLICATIONs

*Co-first authors / †Co-corresponding authors

[International]

Kim, Y., Jeong, S. C., Kim, D.†, & Sim, S.† (2026). Legacy Learning Strategy Based on Few-Shot Font Generation Models for Automatic Text Design. *Information Processing & Management*, 63(6), 104721.

Kim, Y.*, Kim, D.*, Kim, J., Sim, S. (2026). Distributed Lag Transformer based on Time-Variable-Aware Learning for Explainable Multivariate Time Series Forecasting. *IEEE Access,* 14, 115642-115664

Hong, S., Park, T., Kim, D., Jo, S., & Bae, H.† (2026). LURKER: Language Models for Unveiling Maritime Risks via Knowledge-Driven Event Forecasting and Reasoning. *ICIC Express Letters*, 20(5), 543-552.

Jo, S., Kim, D., Hong, S., Park, T., Lee, C., & Bae, H.† (2026). Identifying Severity of Maritime Accidents Based on Weighted Kernel Density Estimation. *ICIC Express Letters,* 20(4). 391-400.

Jo, S.*, Lee, G.*, Park, S., Heo, J., Kim, D., Sim, S.†, & Bae, H.† (2026). Deep learning framework for regional maritime collision risk assessment using CNN and Grad-CAM. *Ocean Engineering*, 343, 123452.

Iman, A. N., Kamal, I. M., Kim, D., & Bae, H.† (2025). Continual Learning Approach With Adaptive Memory Mechanism for Predictive Process Monitoring in Dynamic Business Environments. *IEEE Access,* 13, 206767-206783.

Kim, D., Lee, E., Kamal, I. M., & Bae, H.† (2025). Long-term Forecasting of Maritime Economics Index using Time-series Decomposition and Two-stage Attention. *Journal of forecasting,* 44(1), 153-172.

Iman, A. N., Kamal, I. M., Ramadhan, M. H., Kim, D., Lee, Y., & Bae, H.† (2024). Predictive Process Monitoring for Remaining Time Prediction with Transfer Learning. *ICIC Express Letters*, 18(8), 851-858.

Lee, J.*, Kim, D.*, & Sim, S. (2024) Temporal Multi-Features Representation Learning based Clustering for Time Series Data. *IEEE Access,* 12, 87675-87690.

Kim, D., Lee, S., Sim, S., & Bae, H.† (2024). A Robust eXplainable Feature Selection Method Based on Order Statistics. *ICIC Express Letters,* 18(3), 255-262.

Ramadhan, M. H., Kamal, I. M., Kim, D., & Bae, H.† (2023). Solving the Inter-Terminal Truck Routing Problem for Delay Minimization Using Simulated Annealing with Normalized Exploration Rate. *Journal of Marine Science and Engineering*, 11(11), 2103.

Sim, S.*, Kim, D.*, & Bae, H. (2023). Correlation recurrent units: A novel neural architecture for improving the predictive performance of time-series data. *IEEE Transactions on Pattern Analysis and Machine Intelligence* ,45(12), 14266-14283. *(JCR Top 0.85%).*

Lee, H., Kim, D., Lee, S., Park, H., Bae, H., & Choi, G. (2023). Ensemble Method for Feature Selection in Time-Series Prediction Based on Image Similarity. *ICIC Express Letters Part B: Applications*, 14(7), 709-717.

Utama, N. I., Kamal, I. M., Iman, A. N., Kim, D., Bae, H.†**, & Hong, C.** (2022) Colored Petri Nets simulation on the cloud with python: A framework overview. *ICIC Express Letters, Part B: Applications*, 13(7), 725-733.

Lee, E., Kim, D., & Bae, H.† (2021). Container volume prediction using time-series decomposition with a long short-term memory model. *Applied Sciences*, 11(19), 8995.

Choi, Y., Kwun, H., Kim, D., Lee, E., & Bae, H.† (2021). Residual life prediction for induction furnace by sequential encoder with s-convolutional LSTM. *Processes*, 9(7), 1121.

Lee, E., Park, K., Kim, D., Bae, H., & Hong, C. (2021). Prediction of the quay crane's handling time with external handling factors. *ICIC express letters. Part B, Applications: an international journal of research and surveys*, 12(4), 351-358.

Kamal, I. M., Bae, H., Sim, S., Kim, H., Kim, D., Choi, Y., & Yun, H. (2019). Forecasting High-Dimensional Multivariate Regression of Baltic Dry Index (BDI) using Deep Neural Networks (DNN). *ICIC Express Letters*, 13(5), 427-434.

[Domestic]

Lee, H., Kim, D., Choo, J., Jo, S., & Bae, H.† (2025). A Forecasting Framework of Ship Fuel Consumption using Temporal Patterns in Sensor Data. *Journal of the Korean Institute of Industrial Engineers*, 51(5), 425-438. 10.7232/JKIIE.2025.51.5.425

Han, M. G., Kim, D., & Bae, H.† (2025). Welding Defect Classification Based on Threshold Optimization: A Solution to Class Imbalance Challenges. *Journal of the Korean Institute of Industrial Engineers*, 51(6), 515-526. 10.7232/JKIIE.2025.51.6.515

Kim, M., Heo, J., Kim, D., & Bae, H.† (2024). A Study on Text-Based Weight Extraction for Enhanced SCFI Index Prediction. *The Korean Journal of BigData*, 9(2), 123-134. 10.36498/kbigdt.2024.9.2.123

Park, M., Park, H., Kim, M., Kim, D., Park, Y., & Bae, H.† (2024). Estimation Research for First Prediction Time in Condition-Based Maintenance. *The Journal of Society for e-Business Studies*, 29(1), 93-114. 10.7838/jsebs.2024.29.1.093

Kim, D., Lee, S., Jo, S., Park, S., & Bae, H.† (2024). A Study on the Derivation of Sea Grid and Severity Indicators Based on Maritime Accident Data. *The Journal of Society for e-Business Studies*, 29(1), 57-72.

Park, D., Jo, S., Park, D., Lee, Y., Kim, D., & Bae, H.† (2023). LCL Cargo Loading Algorithm Considering Cargo Characteristics and Load Space. *Journal of Intelligence and Information Systems*, 29(4), 375-393. 10.13088/jiis.2023.29.4.375

Kim, D., Lee, S., & Bae, H.† (2023). A Study on Container Freight Forecasting Model Using eXplainable Artificial Intelligence. *The Asian Journal of Shipping and Logistics*, 39(2), 197-215. 10.37059/tjosal.2023.39.2.197

Yu, T., Lee, Y., Park, H., Kim, D., & Bae, H.* (2022). An Efficient Mixed-Integer Programming Model for Berth Allocation in Bulk Ports. *Journal of the Korea Industrial Information Systems Research*, 27(6), 105-114. 10.9723/jksiis.2022.27.6.105

Kim, D., Han, Y., Kim, H., Kang, S. P., Kim, K. H., & Bae, H.† (2022). Outlier detection and labeling of ship main engine using LSTM-AutoEncoder. *The Korea Journal of Bigdata*, 7(1), 125-137. 10.36498/kbigdt.2022.7.1.125

Kwun, H., Kim, D., Park, C., Lee, L., Cho, G., & Bae, H.† (2020). A Direction of Politic Support for Infectious Disease in Busan Using Time-series Clustering: Focusing on COVID-19 Cases. *The Korea Journal of Bigdata*, 5(1), 125-138. 10.36498/kbigdt.2020.5.1.125

Park, K., Kim, D., Kim, S., Choi, J., & Bae, H.† (2020). Budget Allocation for Emergency Support Funding System During Global Pandemic. *The Korea Journal of Bigdata*, 5(2), 97-110. 10.36498/kbigdt.2020.5.2.97

Kim, D., Kim, H., Sim, S., Choi, Y., Bae, H.†**, & Yun, H.** (2019). Prediction of dry bulk freight index using deep learning. *Journal of Korean Institute of Industrial Engineers*, 45(2), 111-116. 10.7232/JKIIE.2019.45.2.111

CONFERENCEs

Lee, Y., Kim, D., Kim, D., & Bae, H. (2025, August). Multi-task Trained Graph Neural Network for Business Process Anomaly Detection with a Limited Number of Labeled Anomalies. *In International Conference on Business Process Management (pp. 361-378*). Cham: Springer Nature Switzerland.

Kim, D., Park, K., Kim, M., Kwon, J., & Bae, H. (2025). Smart Ports for a Green Future: Integrating Spatiotemporal AI. *Informs International Meeting*. 2025.07.21

Lee, G., Kim, D., Park, S., Sim, S., Liu, L., & Bae, H. (2024, October). Neural Bezier Interpolation with Manifold Learning for Reliable Vessel Trajectory Prediction. *In 2024 IEEE 6th International Conference on Cognitive Machine Intelligence (CogMI)* (pp. 189-196). IEEE.

Park, H., Kim, D., Kim, M., Park, M., Bae, H., & Park, Y. (2023, December). A novel training mechanism for health indicator construction and remaining useful lifetime (RUL) prediction. *In 2023 IEEE International Conference on Big Data (BigData)* (pp. 809-818). IEEE.

Kim, M., Park, H., Park, M., Kwon, J., Kim, D., Bae, H., & Park, Y. (2023). Data-driven degradation trend construction method for RUL prediction. *International Conference on Convergence Content (ICCC) 2023*, 43-44.

Kim, D., Sim, S., Yoon, B., Liu, L., & Bae, H. (2023, November). Improving Time-Series Classification Accuracy Based on Temporal Feature Representation Learning Using CRU-LSTM Autoencoder. *In 2023 IEEE 5th International Conference on Cognitive Machine Intelligence (CogMI)* (pp. 175-183). IEEE.

Kim, D., Park, K., Yu, T., & Bae, H. (2023). A Multi-Objective Reinforcement Learning for Berth Allocation Problem in Tidal Bulk Terminals. *The 11th International Conference on Logistics and Maritime Systems (LOGMS 2023)*. 2023.09.06

Kim, D., Lee, S., Sim, S., & Bae, H. (2023). A Robust eXplainable Feature Selection Method Based on Order Statistics. *The 17th International Conference on Innovative Computing, Information and Control (ICICIC 2023)*. 2023.08.31

Sim, S., Kim, D., & Jeong, S. C. (2023, February). Temporal Attention Gate Network With Temporal Decomposition for Improved Prediction Accuracy of Univariate Time-Series Data. *In 2023 International Conference on Artificial Intelligence in Information and Communication (ICAIIC)* (pp. 122-127). IEEE.

Kim, D., & Bae, H. (2022, December). TDTA: A New Hybrid Framework for Long-term Forecasting of Container Volume. *In 2022 IEEE International Conference on Big Data (Big Data)* (pp. 6718-6720). IEEE.

Choi, Y., Kwun, H., Kim, D., Lee, E., & Bae, H. (2020, February). Method of predictive maintenance for induction furnace based on neural network. *In 2020 IEEE International Conference on Big Data and Smart Computing (BigComp)* (pp. 609-612). IEEE.

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PATENTs

Jul. 2026	**(US patent)** Correlation Recurrent Unit for Improving Prediction Performance of Time-Series Data and Correlation Recurrent Neural Network, No. 12,694,266, 2026.07.28
	(KR patent) No. 10-2916821, 2026.01.20
	(US patent) No. 18/086.646, 2022.12.22
	(KR patent) No. 10-2022-0024013, 2022.02.24
Feb. 2026	**(KR patent)** Method and Apparatus for Fourier-Efficient Adaptive Temporal Hierarchy Forecasting for Time-Series Prediction, No. 10-2026-0030994, 2026.02.19
Jan. 2026	**(KR patent)** Method and Apparatus for Managing Transponder Abnormalities Using Sensor Data of an Automated Guided Vehicle, No. 10-2026-0011178, 2026.01.20
	(KR patent) Method and Apparatus for Maritime Accident Prediction and Control, No. 10-2026-0005507, 2026.01.12
Apr. 2025	**(US patent)** Imputation Method and Device for Missing Values in Dual-Directional Ais Data Based on Deep Learning, No. 19/190,810, 2025.04.28
	(KR patent) No. 10-2024-0057371, 2024.04.30
Feb. 2025	**(KR patent)** Continual Learning Method and Device with Adaptive Memory Mechanism for Predictive Process Monitoring, No. 10-2025-0026041, 2025.02.27
Nov. 2024	**(KR patent)** System and Method for Yard Congestion Prediction and Optimized Truck Arrival Time Guidance, No. 10-2024-0169982, 2024.11.25
Nov. 2024	**(KR patent)** System and Method for Optimizing Yard Operations based on Multi-agent Reinforcement Learning Including External Trucks, No. 10-2024-0169981, 2024.11.25
Sep. 2024	**(KR patent**) Method and Apparatus for Predicting Ship Operation Time, No. 10-2024-0126125,
2024.09.13	
Jun. 2024	**(KR patent)** Apparatus for Prediction of Remaining Useful Lifetime Based on Fluctuation Correlation Recurrent Unit and Method of the Same, No. 10-2024-0081103, 2024.06.21
	(KR patent) Apparatus for Health Indicator Construction Utilizing Semantic Similarity Representation and Method of the Same, No. 10-2024-0081102, 2024.06.21
	(KR patent) Method for Predicting Travel Path of a Vehicle on a Grid-Type Road with Multiple Intersections and Computing Apparatus Thereof, No. 10-2024-0075306, 2024.06.10
Mar. 2024	**(KR patent)** Bayesian Network based Container Dwell-time Prediction Method and Device, No. 10-2024-0034577, 2024.03.12
Sep. 2023	**(US patent)** Text Data-Based Method and System Deducing Social Impact, No. 18/475,214, 2023.09.27
	(KR patent) No. 10-2901286, 2025.12.12
	(KR patent) No. 10-2022-0132276, 2022.10.14
Sep. 2023	**(US patent)** Method and Apparatus for Predicting Recipe Property Reflecting Similarity between Chemical Materials, No. 18/475,206, 2023.09.27
	(KR patent) No. 10-2920277, 2026.01.27
	(KR patent) No. 10-2022-0134746, 2022.10.19
Aug. 2021	**(KR patent)** Apparatus and Method for Predicting IoT-Based Augmented Process, No. 10-2298452, 2021.08.31
Dec. 2020	**(KR patent)** Apparatus and Method for Predicting IoT-Based Augmented Process, No. 10-2020-0180145, 2020.12.21

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PROJECTs

1. "Development of Personalized Smart Driving Logic Based on Customer Usage Patterns Using Process Mining", LG Electronics Inc., Korea, Project Manager, Dec. 2025 – Nov.2026
2. "Human-centered carbon-neutral global supply chain research center", National Research Foundation of Korea (NRF), Korea, Project Manager (International Cooperation, Research), Jun. 2023 – Feb.2026
3. **"**AI-based XPL methodology research**",** National Research Foundation of Korea (NRF), Korea, Project Manager (Management, Research), Mar. 2023 – Nov.2025
4. **"**Meta K-Port intelligent logistics platform", National IT Industry Promotion Agency (NIPA), Korea, Project Manager, May. 2024 – Aug. 2025
5. "Development of CBM+ solution algorithms for key components of radar weapon systems", LIG Nex1 Co., Ltd. Korea, Project Manager, Mar. 2023 – Dec. 2024
6. "Development of indicators for quantitative evaluation of maritime safety", Korea Research Institute of Ships and Ocean Engineering (KRISO), Korea, Researcher, Mar. 2021 – Dec. 2024
7. "Analysis of Busan Port transshipment operations", Goldilocks Shipping Co., Ltd., Korea, Project Manager, Dec. 2024 – Dec. 2024
8. "Block logistics optimization", Samsung Heavy Industries, Korea, Project Manager, Oct. 2023 – Jul. 2024
9. "Analysis of welding defect rates", Samsung Heavy Industries, Korea, Project Investigator, Jun. 2024 – Jul. 2024
10. "Development of big data/AI-based logistics linkage optimization technology", Korea Institute of Marine Science & Technology Promotion (KIMST), Korea, Researcher, Apr. 2022 – Feb. 2024
11. "Establishment of marine and fisheries big data platform and center", National Information Society Agency (NIA), Korea, Project Manager, Sep. 2020 – Dec. 2022
12. "Case analysis of vehicle gate reservation system application in overseas ports", Busan Port Authority (BPA), Korea, Project Manager, Nov. 2022 – Dec. 2022
13. "Development of citrus price and shipment volume forecasting models", Flexink Inc., Korea, Project Manager (Management, Research), Sep. 2022 – Nov. 2022
14. "Development of an AI-based container freight rate prediction model", Korea Maritime Institute (KMI), Korea, Project Manager (Management, Research), Dec. 2021 – Aug. 2022
15. "Industry-academic collaborative research on bulk port operation optimization", CyberLogitec, Korea, Project Manager (Management, Research), Sep. 2021 – Feb. 2022
16. "Design of a service integration framework for real-time interfacing of e-Navi information and port operational status using AI", Korea Institute of Marine Science & Technology Promotion (KIMST), Korea, Researcher, Jan. 2020 – Dec. 2020
17. "Metadata analysis study for operational information display devices for control-based post-management", Game Rating and Administration Committee, Researcher, Korea, Jun. 2020 – Nov. 2020
18. "Process-based simulation analytics for productivity innovation", Korea Technology & Information Promotion Agency for SMEs (TIPA), Korea, Researcher, Dec. 2018 – Nov. 2020
19. "Development of smart port IoT infrastructure technology", Korea Institute of Marine Science & Technology Promotion (KIMST), Korea, Project Manager (Management, Research), May. 2019 – Mar. 2022
20. "Data analysis and prediction model development for Busan Port operational plans (Ship ETA, RTA)", Busan Port Authority (BPA), Korea, Researcher, Dec. 2018 – Aug. 2021
21. "Maritime market analysis using big data", Korea Maritime Institute (KMI), Korea, Project Manager (Management, Research), Dec. 2018 – Aug. 2021